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Constructing new APN functions from known ones

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ABSTRACT

We present a method for constructing new quadratic APN functions from known ones. Applying this method to the Gold power functions we construct an APN function $x^3 + \mathrm{tr}(x^9)$ over $\mathbb{F}_{2^n}$. It is proven that for $n \geqslant 7$ this function is CCZ-inequivalent to the Gold functions, and in the case $n = 7$ it is CCZ-inequivalent to any power mapping (and, therefore, to any APN function belonging to one of the families of APN functions known so far).

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1. Introduction

For any positive integers n and δ, a function F from the field $\mathbb{F}_{2^n}$ to itself is called *differentially δ-uniform* [20] if for every $a \neq 0$ and every b in $\mathbb{F}_{2^n}$, the equation $F(x) + F(x+a) = b$ admits at most δ solutions. Vectorial Boolean functions used as S-boxes in block ciphers must have low differential uniformity to allow high resistance to differential cryptanalysis (see [1]). In this sense, differentially 2-uniform functions, called *almost perfect nonlinear* (APN), are optimal (since for any function, we have $\delta \geqslant 2$).

The notion of APN function is closely connected to the notion of *almost bent* (AB) function [12] which can be described in terms of the *Walsh transform* of functions, that we define now in the framework of the field $\mathbb{F}_{2^n}$ (it can equivalently be defined in the framework of the vector space $\mathbb{F}_2^n$, see

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e.g. [8]). Let $\mathrm{tr}(x)$ denote the trace function from $\mathbb{F}_{2^n}$ into $\mathbb{F}_2$ (that is, $\mathrm{tr}(x) = x + x^2 + x^4 + \cdots + x^{2^{n-1}}$), then the Walsh transform of a given function $F : \mathbb{F}_{2^n} \to \mathbb{F}_{2^n}$ is the integer-valued function

$$\lambda_F(a,b) = \sum_{x \in \mathbb{F}_{2^n}} (-1)^{\mathrm{tr}(bF(x)+ax)}, \quad a, b \in \mathbb{F}_{2^n}.$$

The set $\Lambda_F = \{\lambda_F(a,b)\colon\ a, b \in \mathbb{F}_{2^n}, b \neq 0\}$ is called the *Walsh spectrum* of F and the set $\{|\lambda_F(a,b)|\colon a, b \in \mathbb{F}_{2^n}, b \neq 0\}$ the *extended Walsh spectrum* of F. If the Walsh spectrum of F equals $\{0, \pm 2^{\frac{n+1}{2}}\}$ then the function F is called AB [12]. AB functions exist for n odd only and oppose an optimum resistance to linear cryptanalysis (see [19]). Besides, every AB function is APN [12], and in the n odd case, any quadratic function is APN if and only if it is AB [11]. A comprehensive survey on APN and AB functions can be found in [10].

The APN and AB properties are preserved by some transformations of functions [11,20]. If F is an APN (respectively AB) function, A_1, A_2 are affine permutations and A is affine then the function $F' = A_1 \circ F \circ A_2 + A$ is also APN (respectively AB). The functions F and F' are then called *extended affine equivalent* (EA-equivalent). In [11], Carlet, Charpin and Zinoviev introduced an equivalence relation of functions, more recently called *CCZ-equivalence*, which preserves APN and AB properties: two mappings F and F' from $\mathbb{F}_{2^n}$ to itself are called CCZ-equivalent if for some affine permutation $\mathcal{L}$ of $\mathbb{F}_{2^n}^2$ the image of the graph of F is the graph of F', that is, $\mathcal{L}(G_F) = G_{F'}$ where $G_F = \{(x, F(x)) \mid x \in \mathbb{F}_{2^n}\}$ and $G_{F'} = \{(x, F'(x)) \mid x \in \mathbb{F}_{2^n}\}$. CCZ-equivalent functions have the same differential uniformity and the same extended Walsh spectrum. EA-equivalence is a particular case of CCZ-equivalence and any permutation is CCZ-equivalent to its inverse. In [8], it is proven that CCZ-equivalence is more general.

There are six known classes of power APN functions, four of which are also AB when n is odd (see [16]). It is conjectured by Dobbertin that any power APN function belongs to one of these six classes [15].

Recently a few classes of quadratic APN polynomials CCZ-inequivalent to power functions have been constructed in [2,3,6,7]. Every of these families is defined under a certain condition on n (such as divisibility of n by a certain number) and none of these families is described by polynomials with coefficients in $\mathbb{F}_2$. A natural question is whether all APN polynomials with coefficients in $\mathbb{F}_2$ are CCZ-equivalent to power functions. In the present paper we show that the answer to this question is negative. We give a new approach for constructing quadratic APN functions and using it we construct a class of quadratic APN polynomials with coefficients in $\mathbb{F}_2$. We prove that the function $F(x) = x^3 + \mathrm{tr}(x^9)$ is APN over $\mathbb{F}_{2^n}$ for any n, and that for all $n \geqslant 7$ it is CCZ-inequivalent to the Gold functions (and, hence, EA-inequivalent to all power functions since Gold functions are the only quadratic APN power functions), to the inverse and Dobbertin functions. Obviously, this function is AB for all odd n. We conjecture that for $n \geqslant 7$ the function F is CCZ-inequivalent to any power function. This conjecture is confirmed for the case $n = 7$. Further we show that applying CCZ-equivalence to quadratic APN functions, it is possible to construct classes of nonquadratic APN mappings CCZ-inequivalent to power functions.

2. Construction of new quadratic APN functions

In the theorem below we give a general approach for constructing new quadratic APN functions from known ones.

Theorem 1. *Let F be a quadratic APN function from $\mathbb{F}_{2^n}$ to itself, let f be a quadratic Boolean function on $\mathbb{F}_{2^n}$ and*

$$\varphi_F(x,a) = F(x) + F(x+a) + F(a) + F(0),$$

$$\varphi_f(x,a) = f(x) + f(x+a) + f(a) + f(0).$$

If for every nonzero $a \in \mathbb{F}_{2^n}$ there exists a linear Boolean function ℓ_a satisfying the conditions

(1) $\varphi_f(x,a) = \ell_a(\varphi_F(x,a))$,
(2) *if* $\varphi_F(x,a) = 1$ *for some* $x \in \mathbb{F}_{2^n}$ *then* $\ell_a(1) = 0$,

then the function $F(x) + f(x)$ *is APN.*

Proof. Since the function $F(x) + f(x)$ is quadratic, it is APN if and only if, for every nonzero $a \in \mathbb{F}_{2^n}$, the equation $\varphi_F(x,a) + \varphi_f(x,a) = 0$ admits at most two solutions (see e.g. [10]). According to the hypothesis on ℓ_a, a solution to this equation must be such that $\varphi_f(x,a) = 0$ and therefore such that $\varphi_F(x,a) = 0$. Then, F being quadratic APN, this equation admits at most two solutions. □

The same principle as in Theorem 1 allows generating a large variety of differentially 4-uniform functions from APN functions as it is shown in the proposition below.

Proposition 1. *For any APN function* F *the following functions are differentially* 4*-uniform*

(1) $F(x) + \mathrm{tr}(G(x))$ *for any function* G;
(2) $F \circ A$ *and* $A \circ F$ *for any affine function* A *which is* 2*-to-*1;
(3) *a function* F' *such that* $\mathcal{L}(G_F) = G_{F'}$ *for some affine* 2*-to-*1 *mapping* $\mathcal{L}$ *of* $\mathbb{F}_{2^n}^2$.

Remark 1. Note that, for every APN quadratic function F, every quadratic Boolean function f and every nonzero vector a, a linear function l_a satisfying $\varphi_f(x,a) = \ell_a(\varphi_F(x,a))$ always exists. This is due to the fact that, by the assumption that F is APN, the kernel of $\varphi_F(x,a)$ equals $\{0,a\}$. This set is always a subset of the kernel of $\varphi_f(x,a)$, and this is the necessary and sufficient condition for the existence of l_a.

A direct consequence of Theorem 1 is that, if F is APN and if ℓ is a linear form such that $\ell(1) = 0$, then the function $F(x) + \ell(F(x))$ is APN. But this function is affine equivalent to F since it is equal to $L \circ F$ where $L(x) = x + \ell(x)$, and the condition that $\ell(1) = 0$ is equivalent to saying that L is a permutation.

We give now an example where Theorem 1 leads to a function which is CCZ-inequivalent to the original function F.

Corollary 1. *Let* n *be any positive integer. Then the function* $x^3 + \mathrm{tr}(x^9)$ *is APN on* $\mathbb{F}_{2^n}$.

Proof. We can apply Theorem 1 with $F(x) = x^3$, $\varphi_F(x,a) = a^2x + ax^2$, $f(x) = \mathrm{tr}(x^9)$, $\varphi_f(x,a) = \mathrm{tr}(a^8x + ax^8)$ and $\ell_a(y) = \mathrm{tr}(a^6y + a^3y^2 + a^{-3}y^4)$. Indeed, we have then

$$\ell_a\big(\varphi_F(x,a)\big) = \mathrm{tr}\big(a^6\big(a^2x + ax^2\big) + a^3\big(a^4x^2 + a^2x^4\big) + a^{-3}\big(a^8x^4 + a^4x^8\big)\big) = \varphi_f(x,a)$$

and if there exists $x \in \mathbb{F}_2^n$ such that $\varphi_F(x,a) = 1$ then

$$\ell_a(1) = \mathrm{tr}\big(a^{-3}\big) = \mathrm{tr}\left(\frac{x}{a} + \left(\frac{x}{a}\right)^2\right) = 0. \qquad \square$$

Remark 2. The APN property of the function $x^3 + \mathrm{tr}(x^9)$ can be proven also with the following arguments due to Dillon [14]. If F is a quadratic function then for any nonzero a and for $\varphi_F(x,y) = F(x+y) + F(x) + F(y) + F(0)$ there exists a linear function L_a such that $\varphi_F(ax,a) = L_a(x + x^2)$. Indeed, if $F(x) = \sum_{i \leqslant j} c_{i,j} x^{2^i + 2^j}$ then $L_a(z) = \sum_{i \leqslant j} c_{i,j} a^{2^i + 2^j} (T_{j-i}(z))^{2^i}$, where $T_k(z) = z + z^2 + \cdots + z^{2^{k-1}}$. Thus, F is APN if and only if for any nonzero a and z the equality $L_a(z) = 0$ implies $\mathrm{tr}(z) = 1$. In the case when $F(x) = x^3 + \mathrm{tr}(x^9)$ we have $L_a(z) = a^3z + \mathrm{tr}(a^9T_3(z))$ and if $L_a(z) = 0$ for some $z \neq 0$ then $1 = a^3z = \mathrm{tr}(a^9T_3(z))$ which implies $1 = \mathrm{tr}(z^{-3}(z + z^2 + z^4)) = \mathrm{tr}(z)$.

Another class of APN functions F, to which the construction of Theorem 1 can be applied, is a class of trinomial APN functions described in [6]. However, for this class of functions we were able to construct only functions that are EA-equivalent to the original trinomial F. More precisely we have the following proposition.

Proposition 2. *Let m be a positive odd integer, $n = 2m$, α a primitive element of $\mathbb{F}_{2^n}$. Then the functions $F(x) = x^6 + x^{2^m+1} + \alpha^{2^m-1}x^{6\cdot 2^m}$ and $G(x) = F(x) + \mathrm{tr}(\alpha^{2^{m-1}+1}x^3)$ are EA-equivalent on $\mathbb{F}_{2^n}$.*

Proof. Let $u = \frac{\alpha^{2^{m+1}+1}}{\alpha^{2^m-1}+1}$ and $L(x) = x + \mathrm{tr}(ux)$. It is not difficult to see that $L(F(x)) = G(x)$. □

There is a straightforward generalization of Theorem 1:

Theorem 2. *Let F be a quadratic APN function from $\mathbb{F}_{2^n}$ to itself, let f be a quadratic function from $\mathbb{F}_{2^n}$ to $\mathbb{F}_{2^m}$ where m is a divisor of n, and*

$$\varphi_F(x,a) = F(x) + F(x+a) + F(a) + F(0),$$
$$\varphi_f(x,a) = f(x) + f(x+a) + f(a) + f(0).$$

If for every nonzero $a \in \mathbb{F}_{2^n}$ there exists a linear function ℓ_a from $\mathbb{F}_{2^n}$ to $\mathbb{F}_{2^m}$ which satisfies the conditions

(1) $\varphi_f(x,a) = \ell_a(\varphi_F(x,a))$,
(2) *for every $u \in \mathbb{F}_{2^m}^*$, if $\varphi_F(x,a) = u$ for some $x \in \mathbb{F}_{2^n}$ then $\ell_a(u) \neq u$,*

then the function $F(x) + f(x)$ is APN.

We could find an application of Theorem 2:

Corollary 2. *Let $n = 2m$ where m is an even positive integer. Let us denote by $\mathrm{tr}_{n/m}$ the trace function from $\mathbb{F}_{2^n}$ to $\mathbb{F}_{2^m}$: $\mathrm{tr}_{n/m}(x) = x + x^{2^m}$. The functions $F(x) = x^3 + \mathrm{tr}_{n/m}(x^{2^m+2}) = x^3 + x^{2^m+2} + x^{2^{m+1}+1}$ and $F'(x) = x^3 + (\mathrm{tr}_{n/m}(x))^3$ are APN.*

But unfortunately, these functions are EA-equivalent to power functions. Indeed, let G be the Gold function $G(x) = x^{2^{m-1}+1}$. Let γ be any element of $\mathbb{F}_4 \setminus \mathbb{F}_2$ and L_1, L_2 be the linear mappings $L_1(x) = \gamma^2 x^{2^{m+1}} + \gamma x^2$, $L_2(x) = \gamma x^{2^m} + \gamma^2 x$. Then $G \circ L_1(x) = L_2 \circ F(x)$. Note further that $F'(x) = (F(x))^{2^m}$.

2.1. An algorithmic approach

Below we describe an algorithmic approach to search for functions fulfilling the conditions of Theorem 1 when F is a Gold function. The first step will be to find an explicit description of the linear function l_a used in Theorem 1. Let $F(x) = x^{2^r+1}$ and $f(x) = \mathrm{tr}(\alpha x^{2^i+1})$. Then

$$\varphi_F(x,a) = a^{2^r+1}\left(\left(\frac{x}{a}\right) + \left(\frac{x}{a}\right)^{2^r}\right),$$
$$\varphi_f(x,a) = \mathrm{tr}\left(a^{2^i+1}\alpha\left(\left(\frac{x}{a}\right) + \left(\frac{x}{a}\right)^{2^i}\right)\right).$$

If we define $t = (ir^{-1} - 1) \bmod n$ we get

$$\varphi_f(x,a) = \mathrm{tr}\left(a^{2^i+1}\alpha\left(\sum_{j=0}^{t}\left[\frac{\varphi_F(x,a)}{a^{2^r+1}}\right]^{2^{jr}}\right)\right) = \mathrm{tr}\left(\left(\sum_{j=0}^{t}\alpha^{2^{-jr}}a^{2^{i-jr}+2^{-jr}-(2^r+1)}\right)\varphi_F(x,a)\right).$$

Thus denoting $T_i^r(a,\alpha)=\sum_{j=0}^{t}\alpha^{2^{-jr}}a^{2^{i-jr}+2^{-jr}-(2^r+1)}$ we get

$$\varphi_f(x,a)=\operatorname{tr}\big(T_i^r(a,\alpha)\varphi_F(x,a)\big).$$

In general for $g(x)=\sum_i \alpha_i x^{2^i+1}$ we get

$$\varphi_g(x,a)=\operatorname{tr}\left(\left(\sum_i T_i^r(a,\alpha_i)\right)\varphi_F(x,a)\right).$$

Following Theorem 1, the condition for $F+g$ to be APN is that, if $\operatorname{tr}(a^{-(2^r+1)})=0$ then

$$\operatorname{tr}\left(\sum_i T_i^r(a,\alpha_i)\right)=\sum_i \operatorname{tr}\big(T_i^r(a,\alpha_i)\big)=0.$$

Fixing a base $(b_j)_{j=1,\dots,n}$ of $\mathbb{F}_{2^n}$ over $\mathbb{F}_2$ we can consider the set of vectors

$$\big\{\operatorname{tr}\big(T_i^r(a,b_j)\big)_{a\in\mathbb{F}_{2^n},\operatorname{tr}(a^{-3})=0}\;\big|\;i,j\in\{0\dots n-1\}\big\}.$$

Given F, finding a quadratic function g such that the conditions of Theorem 1 are fulfilled is equivalent to finding a set of linearly dependent vectors in this set. We computed these vectors and all linear dependent sets up to dimension 15. The only examples in addition to $x^3+\operatorname{tr}(x^9)$ are listed below.

(1) For $n=5$ the function $x^5+\operatorname{tr}(x^3)$ is APN (CCZ-equivalent to Gold functions).
(2) For $n=8$ the function $x^9+\operatorname{tr}(x^3)$ is APN (CCZ-inequivalent to power functions and to $x^3+\operatorname{tr}(x^9)$).

3. CCZ-inequivalence of the new APN function to other known APN mappings

Any of the known families of APN functions except the family of the function x^3 cannot cover the family of the function $x^3+\operatorname{tr}(x^9)$. Indeed, by Corollary 1 the function $x^3+\operatorname{tr}(x^9)$ is APN over $\mathbb{F}_{2^n}$ for any n while all previously known APN functions (power mappings as well as polynomials) are defined under some condition on n, except the function x^3 which is APN for any n. Hence, when proving CCZ-inequivalence of $x^3+\operatorname{tr}(x^9)$ to the known APN functions, we mainly concentrate our attention on the Gold functions. Below we prove that $x^3+\operatorname{tr}(x^9)$ is CCZ-inequivalent to Gold mappings for any $n\geqslant 7$. Besides, we prove that for $n=7$ the function $x^3+\operatorname{tr}(x^9)$ is CCZ-inequivalent to any function from the known classes of APN mappings.

Theorem 3. *The function $x^3+\operatorname{tr}(x^9)$ is CCZ-inequivalent to any Gold function on $\mathbb{F}_{2^n}$ when $n\geqslant 7$.*

Proof. Let $F(x)=x^3+\operatorname{tr}(x^9)$ and $G(x)=x^{2^r+1}$ over $\mathbb{F}_{2^n}$ with $n\geqslant 7$, $r\leqslant (n-1)/2$, $\gcd(r,n)=1$. We also assume further that $n\neq 8,10$. For $n=8,10$ CCZ-inequivalence of F and G is checked by a computer.

Suppose first that the functions F and G are EA-equivalent. Then, there exist affine permutations L_1,L_2 and an affine function L' such that

$$L_1\big(x^3\big)+L_1\big(\operatorname{tr}\big(x^9\big)\big)=\big(L_2(x)\big)^{2^t+1}+L'(x).$$

That is,

$$L_1\big(x^3\big)+L_1(1)\operatorname{tr}\big(x^9\big)=\big(L_2(x)\big)^{2^t+1}+L'(x).$$

Since the functions are quadratic, we can assume without loss of generality that L_1 and L_2 are linear: $L_1(x) = \sum_{m\in\mathbb{Z}/n\mathbb{Z}} b_m x^{2^m}$, $L_2(x) = \sum_{p\in\mathbb{Z}/n\mathbb{Z}} c_p x^{2^p}$. Then we get

$$\sum_{m\in\mathbb{Z}/n\mathbb{Z}} b_m x^{3\cdot 2^m} + \operatorname{tr}\big(x^9\big) \sum_{m\in\mathbb{Z}/n\mathbb{Z}} b_m = \sum_{l,p\in\mathbb{Z}/n\mathbb{Z}} c_p c_l^{2^t} x^{2^{l+t}+2^p} + L'(x). \tag{1}$$

On the left-hand side of the identity (1) we have only items of the type $x^{3\cdot 2^m}$, $x^{9\cdot 2^m}$, with some coefficients. Therefore this must be true also for the right-hand side of the identity. Since $n \geqslant 7$ and $n \neq 8, 10$ then there exists a positive integer p different from 1 and 3 such that $\gcd(n, p) = 1$ and $n > 2p$. Since $n > 2p$ then $2^p + 1$ is not in the same cyclotomic coset with 3 or 9. Therefore, the items of the type $x^{2^k(2^p+1)}$ must cancel. That is, for any k

$$c_k c_{k-t+p}^{2^t} = c_{k+p} c_{k-t}^{2^t}. \tag{2}$$

Since $n \geqslant 7$ then 3 and 9 are in different cyclotomic cosets and we have for any k

$$L_1(1) = c_k c_{k-t+3}^{2^t} + c_{k+3} c_{k-t}^{2^t}.$$

If $L_1(1) \neq 0$ then

$$c_k c_{k-t+3}^{2^t} \neq c_{k+3} c_{k-t}^{2^t}. \tag{3}$$

If $c_k \neq 0$ for all k then from (2) and (3) we get

$$c_k c_{k-t}^{-2^t} = c_{k+p} c_{k-t+p}^{-2^t}, \tag{4}$$

$$c_k c_{k-t}^{-2^t} \neq c_{k+3} c_{k-t+3}^{-2^t}. \tag{5}$$

Since $\gcd(n, p) = 1$ and from (4) we obtain $c_k c_{k-t}^{-2^t} = c_m c_{m-t}^{-2^t}$ for any m. It contradicts (5). Thus, $c_k = 0$ for some k. Then from (2) and (3) we get that $c_{k+p} = 0$. Repeating this step for $c_{k+p}, c_{k+2p}, \ldots$ we get $c_{k+ps} = 0$ and since $\gcd(n, p) = 1$ then $c_k = 0$ for all k. A contradiction. If $L_1(1) = 0$ then the equation $L(x) = 0$ has at least 2 solutions $0, 1$ and therefore L_1 is not a permutation. Thus, F and G are EA-inequivalent.

Suppose that $F(x)$ and $G(x)$ are CCZ-equivalent. Then there exists an affine automorphism $L = (L_1, L_2)$ of $\mathbb{F}_{2^n} \times \mathbb{F}_{2^n}$ such that

$$y = F(x) \quad \Leftrightarrow \quad L_2(x, y) = G\big(L_1(x, y)\big)$$

and $L_1(x, F(x))$ is a permutation (see [7]). This implies then

$$L_2\big(x, F(x)\big) = G\big(L_1\big(x, F(x)\big)\big).$$

Writing $L_1(x, y) = L(x) + L'(y)$ and $L_2(x, y) = L''(x) + L'''(y)$ gives

$$L''(x) + L'''\big(F(x)\big) = G\big(L(x) + L'\big(F(x)\big)\big). \tag{6}$$

We can write

$$L(x) = b + \sum_{m\in\mathbb{Z}/n\mathbb{Z}} b_m x^{2^m},$$

$$L'(x) = b' + \sum_{m\in\mathbb{Z}/n\mathbb{Z}} b'_m x^{2^m},$$

$$L''(x) = b'' + \sum_{m\in\mathbb{Z}/n\mathbb{Z}} b''_m x^{2^m},$$

$$L'''(x) = b''' + \sum_{m\in\mathbb{Z}/n\mathbb{Z}} b'''_m x^{2^m},$$

$$b + b' = c.$$

Then we get

$$\begin{aligned}
G\big(L(x)+L'\big(F(x)\big)\big) &= \big(L(x)+L'\big(x^3+\mathrm{tr}\big(x^9\big)\big)\big)\big(L(x)+L'\big(x^3+\mathrm{tr}\big(x^9\big)\big)\big)^{2^r}\\
&= \Bigg[\sum_{m,k\in\mathbb{Z}/n\mathbb{Z}} b_m b_k'^{2^r} x^{2^m+2^{k+r}+2^{k+r+1}} + L'(1)^{2^r}\sum_{m,k\in\mathbb{Z}/n\mathbb{Z}} b_m x^{2^m+2^{k+3}+2^k}\\
&\quad + \sum_{m,k\in\mathbb{Z}/n\mathbb{Z}} b'_m b_k^{2^r} x^{2^{m+1}+2^m+2^{k+r}} + L'(1)\sum_{m,k\in\mathbb{Z}/n\mathbb{Z}} b_m^{2^r} x^{2^{m+r}+2^{k+3}+2^k}\Bigg]\\
&\quad + \Bigg[\sum_{m,k\in\mathbb{Z}/n\mathbb{Z}} b'_m b_k'^{2^r} x^{2^{m+1}+2^m+2^{k+r+1}+2^{k+r}} + L'(1)^{2^r}\sum_{m,k\in\mathbb{Z}/n\mathbb{Z}} b'_m x^{2^{m+1}+2^m+2^{k+3}+2^k}\\
&\quad + L'(1)\sum_{m,k\in\mathbb{Z}/n\mathbb{Z}} b_m'^{2^r} x^{2^{m+r+1}+2^{m+r}+2^{k+3}+2^k}\Bigg] + Q(x),
\end{aligned}$$

where $Q(x)$ is a quadratic polynomial. Obviously, all terms in the expression above whose exponents have 2-weight strictly greater than 2 must cancel.

Since F and G are EA-inequivalent then L' is not a constant. Then there exists $m\in\mathbb{Z}/n\mathbb{Z}$ such that $b'_m\neq 0$.

Let $L'(1)\neq 0$. Since the items with the exponent $2^{m+1}+2^m+2^{m+2}+2^{m+5}$ have to vanish then we get $L'(1)^{2^r}b'_m = L'(1)b_{m-r}'^{2^r}$ and since $L'(1)\neq 0, b'_m\neq 0$ and r is coprime with n then $b'_k\neq 0$ and $b'_k b_{k-r}'^{-2^r} = L'(1)^{1-2^r}$ for all k. Now we can deduce that $b'_{k+r} = L'(1)^{1-2^r}b_k'^{2^r}$ for all k. Then, introducing μ such that $L'(1)^{1-2^r} = \mu^{2^r-1}$, we deduce that $\mu b'_{k+r} = (\mu b'_k)^{2^r}$ for all k and then that $\mu b'_{k+1} = (\mu b'_k)^2$ (using that $\gcd(r,n)=1$) and then $\mu b'_k = (\mu b'_0)^{2^k}$. This means that $\mu L'(x) = \mu b' + \mathrm{tr}(\mu b'_0 x)$. It implies that all nonquadratic items in the last bracket vanish and $L'(x) = d + \mathrm{tr}(d'x)$ for some d, d'. Thus, the function L is not 0 since L' is not a permutation, and then $b_m\neq 0$ for some m. Since the items with the exponent $2^m+2^{m+2}+2^{m+5}$ have to vanish then $L'(1)^{2^r}b_m = L'(1)b_{m-r}^{2^r}$. Like above we get $L(x) = d+\mathrm{tr}(d'x)$. Thus, $L_1(x,F(x)) = d'' + \mathrm{tr}(F'(x))$ for some d'' and $F'(x)$ and $L_1(x,F(x))$ is not a permutation. A contradiction.

Let $L'(1)=0$ and $r\neq 1$. Then $2^{m+1}+2^m+2^{m+r+1}+2^{m+r}$ has 2-weight 4 and since the items with this exponent should cancel then we get $b_m'^{2^r+1} = b'_{m+r}b_{m-r}'^{2^r}$. Since $b'_m\neq 0$ then $b'_{m+r}, b'_{m-r}\neq 0$ and $b'_m b_{m-r}'^{-2^r} = b'_{m+r}b_m'^{-2^r}$. Since $\gcd(n,r)=1$ then $b'_k\neq 0$, $b'_k b_{k-r}'^{-2^r} = b'_m b_{m-r}'^{-2^r}$ for all k and this implies $L'(x) = d+\mathrm{tr}(d'x)$ for some d, d'. Since $L_1(x,F(x))$ is a permutation then $L\neq 0$ and $b_m\neq 0$ for some m. The items with the exponent $2^m+2^{m+r}+2^{m+r+1}$ should vanish. Therefore, $b_m b_m'^{2^r} = b'_{m+r}b_{m-r}^{2^r}$ and

$b_m b_{m-r}^{-2^r} = b'_{m+r} b'^{-2^r}_m$. As above it leads to the equality $L(x) = d + \mathrm{tr}(d'x)$ which is in contradiction with $L_1(x, F(x))$ being a permutation.

Let $L'(1) = 0$ and $r = 1$. Since $L'(1) = 0$ and $b'_m \neq 0$ then there exists t such that $b'_{m+t} \neq 0$. If $t \neq -1, -2$ then $2^{m+1} + 2^m + 2^{m+t+2} + 2^{m+t+1}$ has 2-weight 4 and we get $b'_m b'^{2^r}_{m+t} = b'_{m+t+1} b'^{2^r}_{m-1}$ and $b'_m b'^{-2^r}_{m-1} = b'_{m+t+1} b'^{-2^r}_{m+t}$. Therefore, $L'(x) = d + \mathrm{tr}(d'x)$ for some d, d'. If $t \neq 1, 2$ then $2^{m+t+1} + 2^{m+t} + 2^{m+2} + 2^{m+1}$ has 2-weight 4 and we get $b'_{m+t} b'^{2^r}_m = b'_{m+1} b'^{2^r}_{m+t-1}$ and again $L'(x) = d + \mathrm{tr}(d'x)$ for some d, d'. Thus, $L \neq 0$ and then $b_m \neq 0$ for some m. Since the items with the exponent $2^m + 2^{m+2} + 2^{m+3}$ cancel then $b_m b'^{2^r}_{m+1} = b'_{m+2} b'^{2^r}_{m-1}$ and $b_m b_{m-1}^{-2^r} = b'_{m+2} b'^{-2^r}_{m+1}$. This implies $L(x) = d + \mathrm{tr}(d'x)$ and, thus, $L_1(x, F(x))$ is not a permutation. Therefore, F and G are not CCZ-equivalent. □

Corollary 3. *The function $x^3 + \mathrm{tr}(x^9)$ is EA-inequivalent to any power function on $\mathbb{F}_{2^n}$ when $n \geqslant 7$.*

Proof. The function $x^3 + \mathrm{tr}(x^9)$ is quadratic APN and by Theorem 3 it is EA-inequivalent to any quadratic power function. Since the algebraic degree is EA-invariant then F is EA-inequivalent to any power mapping. □

Dobbertin and inverse APN functions have unique Walsh spectra (except for the case $n = 3$ where the inverse function is EA-equivalent to x^3) which are different from the Walsh spectra of quadratic APN functions (see [9,11,18,21]). Since the extended Walsh spectrum of a function is invariant under CCZ-equivalence, then the function of Corollary 1 is CCZ-inequivalent to the inverse and Dobbertin APN functions for $n \geqslant 7$.

Let $G = \mathbb{F}_2[\mathbb{F}_{2^n} \times \mathbb{F}_{2^n}]$ be the group algebra of $\mathbb{F}_{2^n} \times \mathbb{F}_{2^n}$ over $\mathbb{F}_2$. It consists of the formal sums $\sum_{g \in G} a_g g$ where $a_g \in \mathbb{F}_2$. If S is a subset of $\mathbb{F}_{2^n} \times \mathbb{F}_{2^n}$ then it can be identified with the element $\sum_{s \in S} s$ of G. For any APN mapping F we denote

$$\Delta_F = \big\{(a, b):\ F(x) + F(x + a) = b \text{ has 2 solutions}\big\} \subset \mathbb{F}_{2^n} \times \mathbb{F}_{2^n}.$$

The dimensions of the ideals of G generated by Δ_F and by the graph G_F of F are called Δ- and Γ-ranks, respectively. According to [4,13], Δ- and Γ-ranks of a function are CCZ-invariant.

For $n = 7$ the Δ-rank of the function $F(x) = x^3 + \mathrm{tr}(x^9)$ equals 212 and differs from the Δ-ranks of the Kasami functions x^{13} and x^{23} (which equal 338 and 436, respectively). Thus, for $n = 7$ the function F is CCZ-inequivalent to Kasami functions, and by Theorem 3 to the Gold functions. Since in this field the Welch and Niho cases coincide with the Kasami cases then F is CCZ-inequivalent to all power maps on $\mathbb{F}_{2^7}$. Besides, all previously known families of APN polynomials are not defined for $n = 7$ (see [2,6,7]). Hence we can make the following claim.

Corollary 4. *On $\mathbb{F}_{2^7}$ the function $F(x) = x^3 + \mathrm{tr}(x^9)$ is CCZ-inequivalent to power functions and to any known APN mapping.*

Applying CCZ-equivalence to the quadratic APN function $F(x) = x^3 + \mathrm{tr}(x^9)$, it is possible to construct classes of *nonquadratic* APN mappings which are CCZ-inequivalent to power functions.

For n divisible by 3 we denote $\mathrm{tr}_{n/3}(x) = x + x^{2^3} + x^{2^6} + \cdots + x^{2^{(n/3-1)3}}$ the trace function of $\mathbb{F}_{2^n}$ into $\mathbb{F}_{2^3}$.

Proposition 3. *Let $F : \mathbb{F}_{2^n} \to \mathbb{F}_{2^n}$, $F(x) = x^3 + \mathrm{tr}(x^9)$ then the following functions are CCZ-equivalent to F*

(1) *for n odd the function with algebraic degree* 3

$$x^3 + \mathrm{tr}\big(x^9\big) + \big(x^2 + x\big)\,\mathrm{tr}\big(x^3 + x^9\big);$$

(2) *for n even the function with algebraic degree* 3

$$x^3 + \mathrm{tr}\big(x^9\big) + \big(x^2 + x + 1\big)\,\mathrm{tr}\big(x^3\big);$$

(3) *for n divisible by* 6 *the function with algebraic degree* 4

$$\big(x + \mathrm{tr}_{n/3}\big(x^6 + x^{12}\big) + \mathrm{tr}(x)\,\mathrm{tr}_{n/3}\big(x^3 + x^{12}\big)\big)^3 + \mathrm{tr}\big(\big(x + \mathrm{tr}_{n/3}\big(x^6 + x^{12}\big) + \mathrm{tr}(x)\,\mathrm{tr}_{n/3}\big(x^3 + x^{12}\big)\big)^9\big);$$

(4) *for n odd and divisible by* 3 *the function with algebraic degree* 4

$$\big(x^{\frac{1}{3}} + \mathrm{tr}_{n/3}\big(x + x^4\big)\big)^{-1} + \mathrm{tr}\big(\big(\big(x^{\frac{1}{3}} + \mathrm{tr}_{n/3}\big(x + x^4\big)\big)^{-1}\big)^3\big).$$

Proof. The proof is the same as for the cases from [5,8] (use the affine permutation $\mathcal{L}(x, y) = (x + \mathrm{tr}(y), y)$ for the first two cases, $\mathcal{L}(x, y) = (x + \mathrm{tr}_{n/3}(y^2 + y^4), y)$ for the third case and $\mathcal{L}(x, y) = (x + \mathrm{tr}_{n/3}(y + y^4), y)$ for the fourth case). □

Remark 3. Note that the second and the fourth APN functions in Proposition 3 can be obtained from the functions of [5,8] by adding $\mathrm{tr}(G(x))$ for some G. It means that the construction $F(x) + \mathrm{tr}(G(x))$, where F is APN and G is arbitrary, actually gives new APN functions even if F and G are not quadratic.

Note added in proof: After the appearance of the present paper as a preprint, the idea of adding a Boolean function to a known APN function for obtaining a new APN function has been further used in [17] where an APN function over $\mathbb{F}_{2^6}$ CCZ-inequivalent to power functions and to quadratic functions is found by computer investigations. Finding a method producing an infinite class of such APN functions (in particular, which includes this isolated function) is an open question.

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